

# KENNEDI J. WHITE

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## ◆ EDUCATION

### University of California, Santa Cruz

Expected 2030

PhD in Earth & Planetary Sciences

*Doctoral Advisor: Jonathan Fortney*

### Howard University

Jun. 2021 - May 2025

B.S. with Honors in Physics & Astronomy, B.A. with Honors in English; 3.87 GPA

## ◆ HONORS, AWARDS, & ACHIEVEMENTS

- GEM Fellowship Awardee
- Eugene Cota-Robles Fellowship Awardee
- Future of STEM Scholars Initiative (FOSSI) : scholars selected and awarded to pursue degrees in STEM at Historically Black Colleges and Universities (HBCUs) (2021 - 2025)
- Karsh STEM Scholar : Thirty-two selected scholars committed to earning a PhD or MD/PhD in STEM (2021-2025)

## ◆ EXPERIENCE

### Post-Baccalaureate Research Intern

May 2025 - Aug. 2025

GEM Fellowship Program | Los Alamos National Laboratory

- Identified gamma ray bursts using ground based operations
- Detected gamma ray bursts and modeled their dispersion using trigger algorithms

### Summer Student Research Intern

Jun. 2024 - Aug. 2024

MIT Summer Research Program | Massachusetts Institute of Technology

- Detected directly imaged exoplanets using convolutional neural networks
- Used machine learning to examine parts of an astral image taken by telescopes and coronagraphs to distinguish noise and exoplanets

### Student Research Intern

May 2023 - Present

Maximizing Student Potential in STEM Program | NASA Jet Propulsion Laboratory

- Observation and Analysis of Imaging and Spectroscopy of Outer Planet Atmospheres
- Reduced and analyzed the near infrared thermal images and spectra of Jupiter to understand 2018-2022 EZ disturbance

### Research Assistant

Aug. 2022 - Present

Lunar Environment and Dynamics for Exploration Research | National Aeronautic and Space Administration

- Examining the effect of a solar storm on the Moon's OH (-hydroxyl) surface content and H2 exosphere
- Modeling the lunar exosphere and simulating solar storms to better understand the lunar hydrogen cycle

### Storyteller Apprenticeship Recipient

Aug. 2022 - Jun. 2023

Deep Storyteller Apprenticeship | Savannah, Georgia

- Sponsored podcast created with a peer entitled: Black College Chronicles with Ken and Bri

### Inaugural Cohort Member

Jun. 2022 - Jul. 2022

NASA Student Airborne Science Activation Internship | National Aeronautic and Space Administration

- Summer internship covering topics such as basic remote sensing principles, basic atmospheric thermodynamics, atmospheric stability, and data analysis
- Individual research project on using machine learning ensemble models to measure model accuracy on unseen data titled Summer air quality index prediction with machine learning model classifiers

## A.R.T Member

Jan. 2019 - Jun. 2022

Deep Action Research Team | Savannah, Georgia

- Student representative on Savannah Racial Equity and Leadership (REAL) Task Force (2020-2021)
- Lead facilitator for the creation of the student Bill of Demands presented to city and county council (2020)
- Served as student representative with school board and superintendent to draft new student code of conduct (2020)
- City-wide spoken word performances on toxic masculinity, race, sexuality, etc. (2019 - Feb. 2020)
- Served as student representative with local juvenile judges to draft new student favored legislation (2019)

## ◆ PRESENTATIONS

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### Conference Talks

◆ NASA Exploration Science Forum. July 2023. "Variation of the Moon's Solar-Induced Hydrogen Cycle during a Solar Storm."

### Conference Posters

- ◆ 2024 American Geophysical Union National Conference. December 2024. "Monte Carlo Modeling to Study Variability of the Lunar Solar-Induced Hydrogen Cycle."
- ◆ 2023 American Geophysical Union National Conference. December 2023. "The Moon's Solar-Induced Hydrogen Cycle during Solar Storms."
- ◆ 22nd American Meteorological Society Student Conference. January 2023. "Summer Air Quality Index Prediction with Machine Learning Model Classifiers"

### Other Presentations

- ◆ MIT Summer Research Program. August 2024. "A.I. x Astro: Machine Learning Models for Exoplanet Detection."
- ◆ Howard University Research Symposium. April 2024. "Variation of the Moon's Solar-Induced Hydrogen Cycle during a Solar Storm."
- ◆ Howard University Research Symposium. April 2023. "Impact of Solar Storms on the Moon's Hydrogen Cycle"

## ◆ PUBLICATIONS

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**White, K. J.,** Tucker, O. J., Misra, P. et al. Changes in the Moon's Solar-Induced Hydrogen Cycle During Coronal Mass Ejections

*In Prep*

**White, K. J.,** Dahl, E., Glenn O. et al. Investigating Jupiter's Atmosphere Through Near-Infrared Thermal Imaging and Spectral Analysis

*In Prep*

Tucker, O.J. Farrell, W.M. Li, S. et al. including **White, K. J.** Solar Wind Implantation and Hydrogen Diffusion: Model Comparisons to M3 Observations

<https://doi.org/10.3847/PSJ/ae8e7b>

## ◆ SERVICE & LEADERSHIP

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| <b>President, Corresponding Secretary,</b> Zeta Phi Beta Sorority, Incorporated | <b>2023 - 2025</b> |
| <b>Executive Member, General Member,</b> Society of Physics Students            | <b>2022 - 2025</b> |
| <b>Co-President, Executive Officer,</b> Tau Beta Pi - DC Alpha Chapter          | <b>2022 - 2025</b> |

♦ TECHNICAL SKILLS & EXPERIENCE

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|---------------------------|---------------------------------------------------------------------------|
| Programming               | Python, C++, MATLAB, Fortran, IDL                                         |
| Observatories/Instruments | Lick/Shane & Nickel                                                       |
| Tools                     | LaTeX, Microsoft Office, SAO DS9, Git, GEANT, Fusion 360                  |
| Python Libraries          | numpy, scipy, astropy, pandas, matplotlib,<br>pytorch, pyplanetary-tester |
| Languages                 | English (Native)                                                          |